

Quantum Explainer

An installable, offline-first web app that teaches quantum computing on a real (tiny) simulator - and refuses to hype.



Live (installable PWA): <https://dimitres-kisimov.github.io/quantum-explainer/>

WHAT IT IS

- A hand-written 1-2 qubit state-vector simulator (sim.js, zero dependencies) driving a circuit playground: H/X/Y/Z/S/T, RX/RX/RZ with an angle slider, CNOT, seeded Born-rule measurement, and an explicit entanglement (factorability) check.
- A draggable Bloch sphere - in two-qubit mode it draws each qubit's reduced state, so entanglement appears honestly as both arrows collapsing to the centre of the sphere.
- Phase dials for every complex amplitude, animated gate sweeps along each gate's true rotation axis, and a step scrubber that replays any intermediate state.
- Cited lessons - superposition, measurement, interference, entanglement, Deutsch's algorithm (phase kickback) and Grover's search (amplitude amplification) - plus "What quantum computers are NOT": the parallel-worlds myth, NISQ-era limits, decoherence, error-correction overhead. Eight references, Nielsen & Chuang to Deutsch & Jozsa.
- A real PWA: manifest, precaching service worker, original icons. No frameworks, no build step, no CDNs, no analytics - fully offline after the first visit.

VERIFIED QUALITY (measured while building this PDF)

- 154 physics/behaviour assertions pass in plain Node (node test/sim.test.mjs): $H|0\rangle$ gives 50/50, H.H interference returns $|0\rangle$, the Bell state measures $\{00: 0.5, 11: 0.5\}$ and fails the factorability check, Deutsch's algorithm answers in one query, and Grover's one-iteration search finds the marked item with certainty.
- 86 structural checks pass (node tools/verify.mjs): manifest and icon integrity, service-worker precache completeness, citation presence, and an offline guard proving the app references no external asset of any kind.
- CI re-runs every gate on each push: JS syntax checks, both harnesses, ruff on the Python helper scripts, and an independent offline-guard grep.

THE HONESTY STANCE

- Every physics claim in the app is either demonstrated live on the simulator or cited to a real source. If a sentence sounds like marketing, it doesn't belong.
- The app says out loud what quantum computers cannot do - no "tries every answer at once", no "your encryption is broken", and no invented market figures here either: this is an educational artifact, and its value is that it is correct.

LIMITATIONS (stated, not hidden)

- Two qubits is the deliberate ceiling - the largest system whose full state (4 complex amplitudes) can be shown completely on screen.
- Gates are ideal and noiseless; real hardware decoheres. The lessons say so, but the playground does not model noise.
- A classical simulator demonstrates the rules of quantum mechanics, not a quantum speedup.